

Information: Scenario Compass Initiative Data

The scenario data for this summer school has been created by the Scenario Compass Initiative (SCI). The Scenario Compass provides an ensemble of science-based scenarios for climate futures, curated for consistency with recent global trends and includes scenarios from various models. For more details, see: <https://scenariocompass.org/about>

Please use the data we provide you only for the academic purposes of this summer school. If you want to use any of the data in your future projects, please download the data of your interest from the following link after creating your personal log-in: <https://download.scenariocompass.org/>

For visual inspection of the various scenarios, feel free to use the Scenario Dashboard of SCI: <https://scenariocompass.org/scenario-dashboard>

Please read the next slide for descriptions of the provided scenario data files.

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All data files list the scenarios for the years 2010 – 2100, divided by model, scenario, region, and variable.

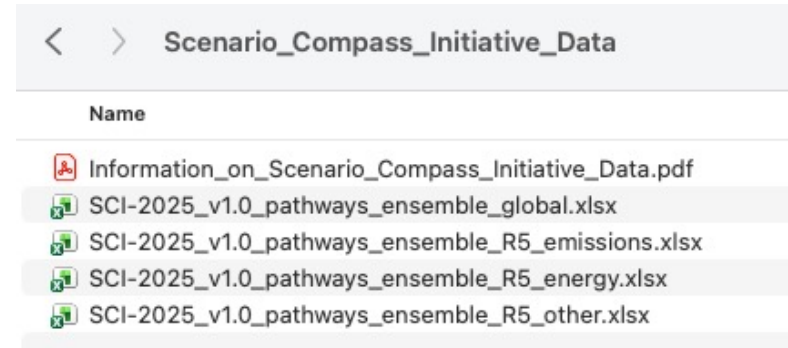
The scenario data divides into four spreadsheet files:

- A global dataset with all scenarios for available models and variables at global level.
- Three regional datasets with all scenarios and variables for five world regions (R5).

The files divide by the variables they include on:

- Emissions
- Energy
- Other

Note that not all models have all variables covered.



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As one example, let's look at the regional data on emissions:

 SCI-2025_v1.0_pathways_ensemble_R5_emissions.xlsx

The data sheet lists all scenario data by different models, scenarios, regions (R5), variables incl. units, from 2010 – 2100:

				D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	
1	Model	Scenario	Region	Variable	Unit	2010	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	2075	2080	2085	2090	2095	2100
2	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions BC	Mt BC/yr	2.99	2.692					2.394					1.871	1.348	1.161	0.975	0.844	0.714	0.612	0.534	0.464	0.405	0.353					
3	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CH4	Mt CH4/yr	150.1	156.8					163.6					108.5	53.43	45.91	38.39	32.84	27.3	22.3	19.71	17.45	16	14.07					
4	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CH4 AFOLU	Mt CH4/yr	79.6	82.94					86.28					62.54	38.8	33.66	28.52	24.51	20.49	17.44	15.89	14.43	13.48	12.07					
5	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CH4 Energy	Mt CH4/yr	44.08	44.63					45.17					26.6	8.03	6.93	5.83	5.165	4.501	3.167	2.404	1.818	1.428	1.085					
6	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO	Mt CO/yr	318	298.2					278.5					241	203.5	184.3	165.2	149.2	133.1	119.2	107.1	95.6	85.04	75.11					
7	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2	Mt CO2/yr	11635	12350					13065					9758	6451	4753	3054	2053	1052	-116.2	-556.8	-754.3	-841.6	-916.7					
8	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 AFOLU	Mt CO2/yr	1626	1450					1275					797.1	319.6	51.15	-217.3	-422.1	-626.9	-755	-816.3	-836.6	-833.6	-812.7					
9	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy and Industrial Processes	Mt CO2/yr	10009	10900					11791					8961	6132	4702	3271	2475	1679	638.9	259.5	82.3	-7.935	-103.9					
10	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy Demand	Mt CO2/yr	4682	4878					5074					4054	3033	2619	2204	1884	1564	1038	869	758.1	704.6	636.2					
11	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy Demand Industry	Mt CO2/yr	3019	3046					3073					2346	1619	1406	1193	1043	893.2	699.4	641.2	594.3	566.6	539.5					
12	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy Demand Residential and Commercial	Mt CO2/yr	841.3	882.3					923.3					751.8	580.4	449.9	319.3	251.9	184.5	72.88	38.49	21.21	15.74	9.638					
13	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy Supply	Mt CO2/yr	5327	6022					6716					4907	3099	2083	1067	590.7	114.5	-399.5	-609.5	-675.8	-712.6	-740.1					
14	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions CO2 Energy Supply Electricity	Mt CO2/yr	4082	4771					5460					3910	2361	1537	712.9	345.5	-21.84	-444.5	-630.4	-682.7	-715.6	-738.1					
15	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions F-Gases	Mt CO2-equiv/yr	127.4	174.2					221					188.6	156.2	99.61	43.03	42.83	42.64	39.31	35.8	32.01	28.01	23.57					
16	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions Kyoto Gases	Mt CO2-equiv/yr	16509	17500					18490					13526	8561	6596	4631	3478	2325	971.4	425.1	128.5	-31.9	-200.8					
17	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions N2O	kt N2O/yr	3341	3539					3738					2906	2074	2001	1927	1882	1837	1647	1521	1392	1281	1143					
18	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions N2O AFOLU	kt N2O/yr	2492	2633					2774					2263	1753	1728	1703	1685	1666	1530	1430	1320	1221	1095					
19	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions NH3	Mt NH3/yr	24.41	25.39					26.37					26.24	26.12	26.13	26.14	26.14	26.14	25.22	24.25	23.08	21.92	20.4					
20	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions NOx	Mt NO2/yr	34.67	33.29					31.92					23.42	14.92	12.3	9.691	8.356	7.022	5.8	5.085	4.459	3.941	3.431					
21	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions OC	Mt OC/yr	11.78	10.8					9.816					8.308	6.801	6.019	5.237	4.643	4.048	3.667	3.356	3.058	2.787	2.539					
22	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions Sulfur	Mt SO2/yr	38.28	34.9					31.52					21.14	10.77	8.146	5.52	4.24	2.959	2.313	1.969	1.675	1.433	1.198					
23	AIM/CGE 2.0	SSP1-19	Asia (R5)	Emissions VOC	Mt VOC/yr	68.26	65.39					62.52					56.12	49.73	46.6	43.47	41.04	38.61	32.44	27.11	21.92	17	12.28					
24	AIM/CGE 2.0	SSP1-19	Latin America (R5)	Emissions BC	Mt BC/yr	0.855	0.779					0.704					0.604	0.505	0.47	0.436	0.411	0.387	0.357	0.33	0.307	0.286	0.266					
25	AIM/CGE 2.0	SSP1-19	Latin America (R5)	Emissions CH4	Mt CH4/yr	48.48	50.52					52.56					36.86	21.15	18.28	15.4	13.11	10.82	9.046	8.124	7.294	6.744	5.938					
26	AIM/CGE 2.0	SSP1-19	Latin America (R5)	Emissions CH4 AFOLU	Mt CH4/yr	33.89	35.39					36.9					27.61	18.31	15.97	13.63	11.68	9.726	8.274	7.488	6.76	6.26	5.524					
27	AIM/CGE 2.0	SSP1-19	Latin America (R5)	Emissions CH4 Energy	Mt CH4/yr	7.782	7.756					7.729					4.431	1.133	0.95	0.767	0.653	0.54	0.363	0.285	0.23	0.203	0.174					

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- Emission-related variables include annual emissions of a specific gas, e.g., CO₂, either across all sectors or divided by demand/supply in different sectors.
- Energy-related variables cover final, primary, and secondary energy, divided by energy sectors (e.g., industry, transport), and technologies/fuels (e.g., oil, gas, hydro, solar, wind).
- Other variables include, for instance, agriculture, population, technology capacities, land cover.